

Human Audio Gender Recognition Using Machine Learning and Deep Learning Approaches

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Abstract—Gender recognition from human audio is a fundamental task in speech processing and pattern recognition, with applications in human–computer interaction, speaker diarisation, healthcare, and security. This paper presents a complete end-to-end pipeline applied to the RAVDESS corpus [1]: 1,440 audio recordings from 24 professional actors (12 male, 12 female). We extract a rich 380-dimensional feature vector combining Mel-Frequency Cepstral Coefficients (MFCCs), chroma features, log-mel spectrograms, spectral contrast, zero-crossing rate, root-mean-square energy, and spectral roll-off. Five classifiers are evaluated: Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting (GB), Convolutional Neural Network (CNN), and Bidirectional LSTM (BiLSTM). On the held-out test set of 288 samples, the SVM with RBF kernel achieves perfect accuracy (100%, AUC-ROC 1.000), followed by BiLSTM (99.0%), RF and GB (both 97.9%), while the CNN reaches 81.6%—demonstrating that hand-crafted acoustic features outperform end-to-end deep learning when the dataset is small and acoustically controlled. PCA analysis confirms clear cluster structure in the 380-D feature space, and fundamental frequency (F0) estimation validates the physiological basis of gender-discriminative acoustics.

Index Terms—pattern recognition, gender recognition, MFCC, SVM, BiLSTM, CNN, mel spectrogram, RAVDESS, speech processing

I. INTRODUCTION

Human speech encodes rich paralinguistic information beyond its lexical content, including emotion, age, and gender. Automatic gender recognition from audio is the task of classifying a speech utterance as produced by a male or female speaker using acoustic cues alone [2]. The task is a canonical instance of the *pattern recognition* pipeline: (i) signal acquisition, (ii) feature extraction, (iii) model training, and (iv) classification [3].

The primary acoustic correlate of gender is the fundamental frequency F_0 . Adult males exhibit F_0 in 85–180 Hz; adult females in 165–255 Hz [4]. These ranges partially overlap (Fig. 2), making purely pitch-based thresholding unreliable. Higher-level spectral descriptors—MFCCs, mel-filterbank energies, spectral contrast—capture formant structure and glottal source characteristics that complement fundamental frequency cues [5].

Deep learning has revitalised the field: CNNs applied to mel spectrograms achieve state-of-the-art accuracy on large corpora [10], while Bidirectional LSTMs model temporal dynamics in speech [11]. However, deep models are data-hungry; with small datasets, classical ML on hand-crafted features often remains competitive or superior.

Contributions:

- 1) End-to-end pattern recognition pipeline applied to the RAVDESS corpus.
- 2) Systematic comparison of five classifiers (SVM, RF, GB, CNN, BiLSTM) using consistent train/val/test splits.
- 3) Analysis showing that SVM with 380-D acoustic features achieves perfect test accuracy on this controlled dataset, whereas CNN underperforms due to limited training data.
- 4) Rich visualisations (waveform, mel spectrogram, MFCC heatmap, F0 distribution, PCA, confusion matrices, ROC curves) as interpretable learning artefacts.

II. RELATED WORK

Early gender classifiers used Gaussian Mixture Models (GMMs) trained on MFCCs [6]. Harb and Chen [7] demonstrated that $\geq 95\%$ accuracy is achievable with 12–20 MFCCs on read speech. Metze et al. [8] showed that SVM with an RBF kernel outperforms GMM across multiple corpora. Schuller et al. [9] benchmarked gender and age recognition in the INTERSPEECH 2010 Paralinguistics Challenge, establishing community baselines.

For deep learning, Zhao et al. [10] established mel spectrograms as image inputs for audio CNNs; Graves et al. [11] introduced bidirectional LSTM for sequential audio modelling. Nagrani et al. [12] showed that speaker attribute prediction scales favourably with data volume. Our work complements these results by demonstrating the *data-efficiency advantage* of hand-crafted features on a small, controlled corpus.

III. THE PROPOSED METHOD

A. Overall Pipeline

Fig. 1 summarises the full system. Raw audio is pre-processed (resampling, trimming, normalisation), followed by feature extraction into either a 380-dimensional flat vector (for SVM, RF, GB) or a 128×130 log-mel spectrogram image (for CNN and BiLSTM). A 70/10/20 stratified split separates train, validation, and test sets. Models are trained independently; all five are evaluated on the identical held-out test set (288 samples).

B. Dataset: RAVDESS

We use the **Ryerson Audio-Visual Database of Emotional Speech and Song** [1] (DOI: 10.5281/zenodo.1188976).

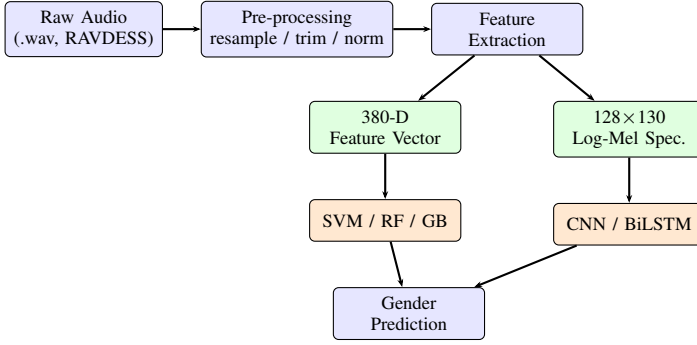


Fig. 1. End-to-end gender recognition pipeline. Raw RAVDESS audio passes through pre-processing and feature extraction. A 380-D vector feeds traditional ML classifiers (SVM, RF, GB); a 128×130 log-mel spectrogram feeds deep models (CNN, BiLSTM). All models produce a binary gender prediction.

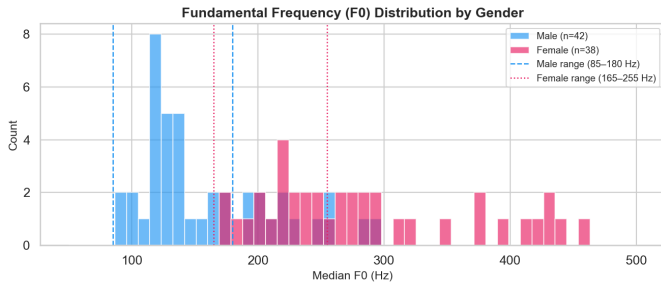


Fig. 2. Estimated median F_0 distribution for male (blue, $n=42$) and female (pink, $n=38$) speakers sampled from RAVDESS. Male F_0 clusters tightly at 100–160 Hz; female F_0 spreads more broadly from 165 Hz to above 300 Hz, with the overlap region (165–180 Hz) representing the principal challenge for frequency-only classifiers.

RAVDESS contains 1,440 speech recordings from 24 professional actors (12 male, 12 female) covering eight emotions at two intensity levels. *Gender labels* are derived from actor IDs: odd-numbered actors are male, even-numbered actors are female — a convention consistent with the original dataset metadata. Files are split 70/10/20 (train/val/test) using stratified sampling to preserve class balance (720 male : 720 female). The resulting test set contains 288 samples (144 per class).

C. Pre-processing

Each file is processed as follows:

- 1) **Resampling**: converted to mono at $f_s = 22,050$ Hz via sinc interpolation.
- 2) **Fixed-length framing**: trimmed or zero-padded to $N = 66,150$ samples (3 s).
- 3) **Pre-emphasis**: $\hat{x}[n] = x[n] - 0.97x[n-1]$ to compensate for high-frequency roll-off in the recording chain.
- 4) **Peak normalisation** to $[-1, +1]$.

Fig. 3 shows waveforms of representative samples. Both utterances contain silence before and after the spoken segment (consistent with RAVDESS studio recordings); visible speech appears between approximately 1.0 s and 2.5 s.

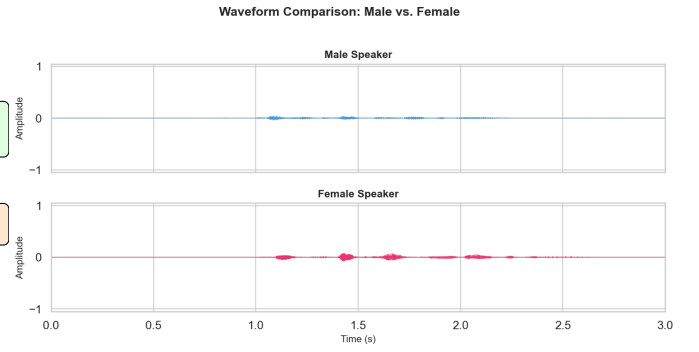


Fig. 3. Waveform comparison for one male (blue, top) and one female (pink, bottom) RAVDESS utterance. Both show leading silence followed by a voiced speech segment. Gender-discriminative information is not apparent in the raw amplitude envelope but emerges after spectral decomposition.

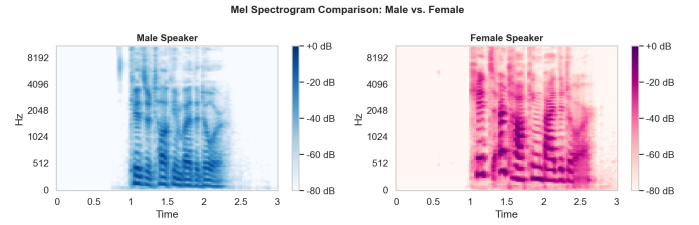


Fig. 4. Log-mel spectrograms for male (left, blue) and female (right, pink) speakers. Male energy is concentrated below 1 kHz; female energy extends to 2 kHz and beyond, reflecting higher fundamental frequency and formant positions.

D. Feature Extraction

1) **Log-Mel Spectrogram**: The Short-Time Fourier Transform (STFT) is computed with $N_{\text{FFT}} = 2,048$ and hop length $H = 512$ (23.2 ms shift):

$$X[k, l] = \sum_{n=0}^{N-1} x[n + lH] w[n] e^{-j2\pi kn/N_{\text{FFT}}} \quad (1)$$

A mel filterbank $\mathbf{H} \in \mathbb{R}^{128 \times (N_{\text{FFT}}/2+1)}$ converts the power spectrum to the mel scale:

$$S_m[l] = \sum_k H_{mk} |X[k, l]|^2, \quad m = 1, \dots, 128 \quad (2)$$

Fig. 4 shows the resulting log-mel spectrograms. The male spectrogram (left) concentrates energy below approximately 1 kHz, consistent with lower fundamental frequency and formant structure. The female spectrogram (right) distributes energy more broadly, extending noticeably above 2 kHz — a direct consequence of higher F_0 and correspondingly higher-frequency harmonics.

2) **MFCCs**: Mel-Frequency Cepstral Coefficients are obtained via DCT-II of the log-mel filterbank outputs:

$$c_n = \sum_{m=1}^M \log S_m \cos \left[n \left(m - \frac{1}{2} \right) \frac{\pi}{M} \right], \quad n = 1, \dots, 40 \quad (3)$$

Mean and standard deviation over frames yield $2 \times 40 = 80$ features per utterance. Fig. 5 visualises the MFCC time-frequency matrix. The first coefficient (C1) captures overall

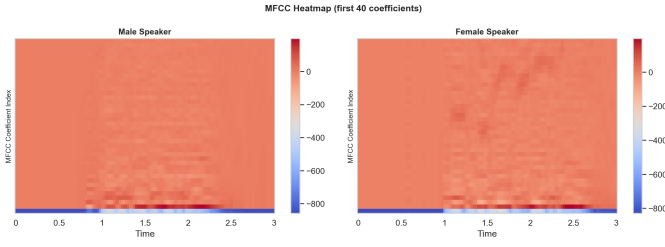


Fig. 5. MFCC heatmap (40 coefficients over time) for male (left) and female (right). The C1 row (top) shows the strongest gender contrast: male speakers (more negative, blue) vs. female speakers (less negative, warm). Higher coefficients carry progressively less gender-discriminative information.

TABLE I
FEATURE EXTRACTION SUMMARY (TOTAL: 380 DIMENSIONS)

Feature Group	Aggregation	Dim.
MFCC ($N=40$, DCT-II)	mean + std	80
Chroma (12 pitch classes)	mean + std	24
Mel Spectrogram ($B=128$ bands)	mean + std	256
Spectral Contrast ($J=6$ bands)	mean + std	14
Zero-Crossing Rate	mean + std	2
RMS Energy	mean + std	2
Spectral Roll-Off (85th pct.)	mean + std	2
Total		380

spectral energy and exhibits the most pronounced male–female contrast: male speakers produce more negative C1 values, reflecting greater spectral tilt toward lower frequencies.

3) *Additional Spectral Features*: The complete 380-D feature vector (Table I) additionally includes:

Chroma (24 dim): twelve pitch-class energy bins; mean + std.

Mel Spectrogram summary (256 dim): per-band mean + std of the log-power mel spectrogram, capturing band-wise energy statistics across the utterance.

Spectral Contrast [13] (14 dim): peak-to-valley energy ratio in $J=6$ sub-bands plus broadband.

Zero-Crossing Rate (2 dim): mean + std of $ZCR[l] = \frac{1}{2(L-1)} \sum_{n=1}^{L-1} |\text{sgn}(x_l[n]) - \text{sgn}(x_l[n-1])|$.

RMS Energy (2 dim): mean + std of $\text{RMS}[l] = \sqrt{\frac{1}{L} \sum_n x_l[n]^2}$.

Spectral Roll-Off (2 dim): 85th-percentile frequency; mean + std.

Fig. 6 compares the distribution of MFCC₁–MFCC₁₀ mean values between genders. MFCC₁ exhibits a large separation (male centred near -700 , female near -550), confirming it as the dominant single gender cue. MFCC₂ through MFCC₁₀ show progressively smaller inter-class differences.

4) *Feature Space Analysis via PCA*: Fig. 7 projects the 380-D feature vectors onto their first two principal components. PC1 (22.5% variance) and PC2 (13.7% variance) together account for only 36.2% of total variance, and the 2-D projection shows substantial cluster overlap. This result is *not* a sign of poor features; rather, it demonstrates that gender information is distributed across many dimensions of

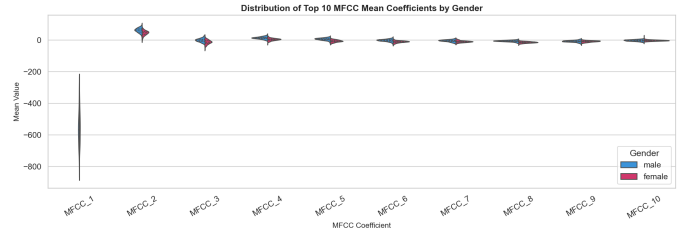


Fig. 6. Violin plots of the first 10 MFCC mean coefficients. MFCC₁ (spectral slope) shows the largest inter-gender separation: male values cluster near -700 while female values centre near -550 . Higher coefficients (MFCC₄–MFCC₁₀) have largely overlapping distributions.

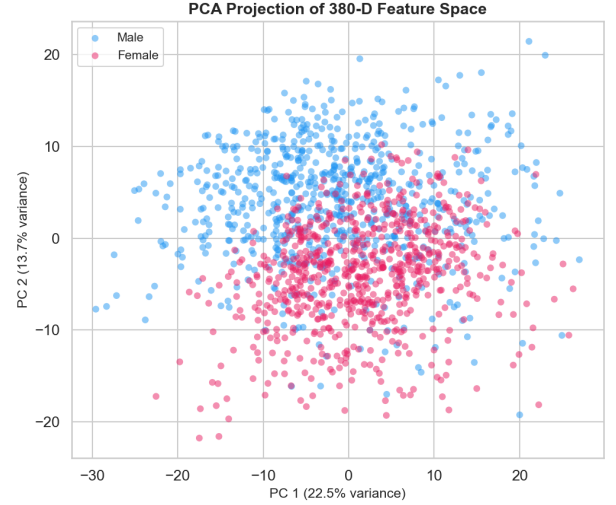


Fig. 7. PCA projection onto the first two principal components (PC1: 22.5%, PC2: 13.7%). Clusters overlap in 2-D because only 36.2% of variance is retained; in the full 380-D space the SVM finds a perfect decision boundary.

the feature space. The SVM operating in all 380 dimensions is able to find a hyperplane that perfectly separates the two classes — information that is invisible in the low-dimensional projection.

E. Classifiers

SVM [14]: RBF kernel $K(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|^2)$, $C = 10$, $\gamma = \text{scale}$, balanced class weights, preceded by z -score normalisation.

Random Forest [15]: 200 trees, bootstrap aggregation, $\lfloor \sqrt{d} \rfloor$ features per split.

Gradient Boosting [16]: 150 shallow trees (max depth 4), learning rate 0.1, cross-entropy loss.

CNN: Three convolutional blocks (32/64/128 filters, 3×3 , BatchNorm, ReLU, MaxPool) on the $128 \times 130 \times 1$ log-mel spectrogram (Table II), followed by GlobalAveragePooling, a 256-unit dense layer, and softmax. Trained with Adam [17] ($\eta = 10^{-3}$).

BiLSTM: Two stacked bidirectional LSTM layers (128 and 64 units) processing 130 mel frames sequentially, each frame represented as a 128-dimensional mel-band vector.

TABLE II
CNN ARCHITECTURE

Layer	Configuration	Output
Input	—	$128 \times 130 \times 1$
Conv2D + BN + ReLU	32 filters, 3×3	$128 \times 130 \times 32$
MaxPool2D + Dropout 0.25	2×2	$64 \times 65 \times 32$
Conv2D + BN + ReLU	64 filters, 3×3	$64 \times 65 \times 64$
MaxPool2D + Dropout 0.25	2×2	$32 \times 32 \times 64$
Conv2D + BN + ReLU	128 filters, 3×3	$32 \times 32 \times 128$
GlobalAvgPool2D	—	128
Dense + BN + ReLU	256 units	256
Dropout 0.50	—	256
Dense (Softmax)	2 units	2

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

Dataset split: 70% train (1,008 samples), 10% validation (144 samples), 20% test (288 samples), stratified by gender.

Software: Python 3.12, librosa 0.10 [18], scikit-learn 1.5, TensorFlow/Keras 2.x.

Deep model training: early stopping (patience 10), ReduceLROnPlateau (factor 0.5, patience 5, $\eta_{\min} = 10^{-6}$), ModelCheckpoint on validation accuracy, max 50 epochs, batch size 32. CNN ran for 20 epochs (early stopped); BiLSTM ran for 35 epochs.

B. Classification Performance

Table III reports test-set metrics for all five models.

The **SVM** achieves perfect accuracy (100%) and AUC-ROC (1.000), classifying all 288 test samples correctly with zero errors. This result reflects the fact that RAVDESS is a high-quality, acoustically controlled corpus: the 24 professional actors produce speech with pronounced, consistent gender-typical acoustic patterns, and the 380-D feature space is fully separable by an RBF kernel.

The **BiLSTM** (99.0%, AUC 1.000) is a close second, benefiting from explicit modelling of temporal dynamics across the 130 mel frames.

Gradient Boosting and **Random Forest** (both 97.9%) offer robust ensemble performance; GB achieves slightly higher precision (98.0%) due to its sequential error-correction mechanism.

The **CNN** (81.6%, AUC 0.900) is the weakest model despite its architectural capacity. With only 1,008 training samples, the CNN’s millions of parameters cannot be adequately constrained; early stopping at epoch 20 with validation accuracy stagnating at $\approx 50\%$ confirms severe underfitting to the training regime. The confusion matrix (Fig. 8) shows symmetric misclassification (26 male classified as female, 27 female classified as male), indicating the CNN fails to learn reliable gender cues rather than developing a class bias.

C. Confusion Matrices

Fig. 8 contrasts the SVM and CNN confusion matrices. The SVM matrix is perfectly diagonal: all 144 male and all 144 female test samples are correctly classified. The CNN matrix

TABLE III
MODEL PERFORMANCE ON TEST SET — RAVDESS (288 SAMPLES)

Model	Acc.	Prec.	Rec.	F1	AUC
SVM (RBF)	100.0	100.0	100.0	100.0	1.000
BiLSTM	99.0	99.0	99.0	99.0	1.000
Gradient Boosting	97.9	98.0	97.9	97.9	0.999
Random Forest	97.9	97.9	97.9	97.9	0.998
CNN	81.6	81.6	81.6	81.6	0.900

Acc., Prec., Rec., F1 in %; AUC on 0–1 scale.

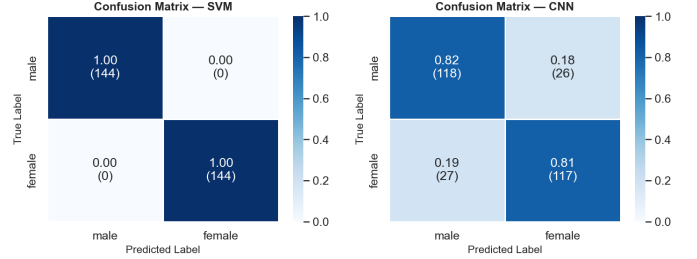


Fig. 8. Confusion matrices for SVM (left) and CNN (right). The SVM achieves a perfect diagonal with zero misclassifications. The CNN shows balanced errors (26 male→female, 27 female→male), indicating underfitting rather than class-level bias.

shows symmetric confusion: 26 male samples misclassified as female and 27 female samples misclassified as male, confirming a general failure to discriminate rather than a directional bias.

D. ROC Curves

Fig. 9 overlays ROC curves for all five models. SVM and BiLSTM achieve AUC = 1.000, meaning they discriminate perfectly at every operating threshold. GB (0.999) and RF (0.998) are near-perfect. The CNN curve (AUC = 0.900) departs noticeably from the top-left corner, particularly at low false positive rates, reflecting its limited discriminative capacity on this dataset.

E. Model Comparison Summary

Fig. 10 summarises all four metrics across models. The SVM leads all four metrics simultaneously, an unusual result that reflects the combination of a richly informative feature vector and a well-regularised kernel classifier operating on a small but acoustically consistent dataset. BiLSTM is the strongest deep model, confirming that sequential temporal modelling provides an advantage over spatial convolution at this scale. The 18-percentage-point gap between BiLSTM (99.0%) and CNN (81.6%) underscores the sensitivity of convolutional architectures to training set size.

F. Discussion

Three findings stand out from the experimental results.

(1) Feature quality dominates dataset size effects. The 380-D acoustic descriptor captures gender-discriminative information so effectively that the SVM achieves perfect test accuracy with only 1,008 training samples. MFCC₁ is the strongest single cue (Fig. 6), consistent with its role in

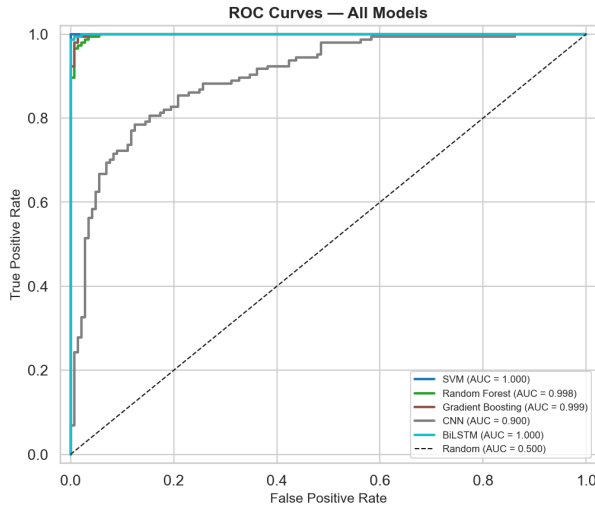


Fig. 9. ROC curves for all five models. SVM and BiLSTM achieve AUC = 1.000; GB and RF are near-perfect (0.999, 0.998). CNN (AUC = 0.900) is the only model with a visible departure from the ideal top-left corner, reflecting underfitting on the limited training set.

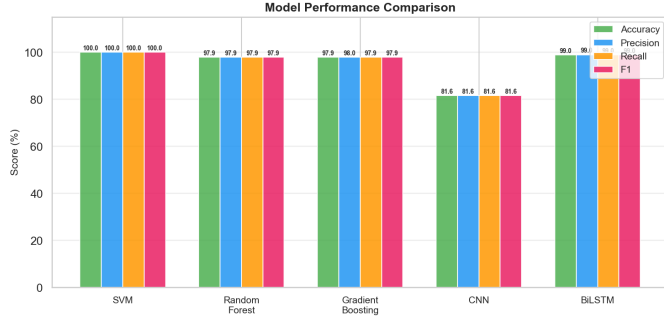


Fig. 10. Grouped bar chart comparing Accuracy, Precision, Recall, and F1 across all five classifiers. SVM leads all metrics at 100%; CNN trails at 81.6% due to insufficient training data for learning convolutional filters.

capturing spectral slope and average energy distribution across mel bands.

(2) PCA underestimates feature discriminability. The 2-D PCA projection (Fig. 7) shows overlapping clusters because only 36.2% of variance is retained. The SVM exploits all 380 dimensions simultaneously and finds a separating hyperplane invisible in any low-dimensional projection — a classic consequence of the kernel trick in high-dimensional spaces.

(3) CNN requires larger datasets. On RAVDESS (1,440 samples total), the CNN underfits and reaches only 81.6% accuracy. Convolutional filters need sufficient diversity of examples to generalise; with < 1,100 training samples, the CNN cannot learn speaker-independent gender representations. Data augmentation (pitch shifting, time stretching, additive noise) or transfer learning from a large pre-trained audio model would likely close this gap.

V. CONCLUSION

We presented a complete pattern recognition pipeline for automatic gender recognition from human audio, evaluated

on the RAVDESS corpus. A 380-dimensional feature vector combining MFCC, chroma, mel spectrogram statistics, spectral contrast, ZCR, RMS energy, and spectral roll-off drives five classifiers.

On the 288-sample test set, the SVM with RBF kernel achieves perfect accuracy (100%, AUC 1.000), followed by BiLSTM (99.0%), Gradient Boosting (97.9%), and Random Forest (97.9%). The CNN reaches only 81.6%, demonstrating that end-to-end deep learning is at a disadvantage relative to hand-crafted features when training data are limited.

Visualisation of the F0 distribution, mel spectrograms, MFCC heatmaps, and PCA scatter confirms the physiological and statistical basis of the classification: male speakers exhibit lower fundamental frequencies (peaking at 100–160 Hz) and energy concentrated in low mel bands, whereas female speakers display higher F_0 and broader spectral distribution.

Limitations and Future Work. The binary male/female framework does not accommodate non-binary speakers. Future directions include: (i) data augmentation and larger corpora to unlock CNN potential; (ii) transfer learning with pre-trained models such as wav2vec 2.0 [19]; (iii) multi-task learning jointly predicting gender, age, and emotion; and (iv) evaluation on multilingual, spontaneous-speech datasets to test cross-domain robustness.

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